PHISHING EMAIL DETECTION & AWARENESS

Task 2 – Cyber Security Internship Project

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Project scope	Email sample review, header interpretation, link/attachment inspection, risk classification and awareness guidance
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Educational analysis only. The examples used in this report are anonymized public training samples. No live phishing link was opened, no credentials were entered, and no malware or attachment was executed.

1. Executive Summary

This report reviews two educational phishing email samples: a fake fax notification and a fake voicemail notification. The review focuses on sender and message context, the supplied header evidence, suspicious links or attachments, and practical indicators that a recipient can use to identify phishing.

Overall assessment: Both samples contain multiple social-engineering warning signs and are classified as **High Risk – Phishing** for training purposes. This is a sample-based assessment, not a live reputation verdict about any real sender or domain.

2. Objectives and Scope

The objectives were to: (1) inspect the provided email content and available header fields; (2) identify sender, authentication, link and attachment warning signs; (3) assign a risk classification; (4) document the findings; and (5) provide prevention and employee-awareness guidance.

Scope limitations: the samples use placeholder domains such as **attacker.example.com** and **target.example.com**, and the IP address **192.0.2.1**. These are anonymized/example values, not verified real infrastructure. Some header fields were unavailable or redacted. No live site or attachment was executed.

3. Methodology and Tools

The review used a manual, read-only workflow:

• Read the sample email body and note the sender, subject, recipient and requested action.
• Review the header fields supplied with each sample, including SPF, DKIM and DMARC results where present.
• Inspect the displayed link destination or attachment name as text; do not click or execute it.
• Record suspicious indicators and assign a qualitative risk level.
• Prepare awareness recommendations based on the observed patterns.

Reference tools suitable for safe header review include Google Admin Toolbox – Messageheader and MXToolbox – Email Header Analyzer. Use these only with sample headers and avoid uploading confidential business or personal email headers to public services without authorization.

4. Sample 1 – Fake Fax Notification

Sample description

The sample presents itself as a fax notification and uses wording similar to “You have a new fax” with a “Print & Preview Here” link. The supplied sender is **attack@attacker.example.com**. The recipient field is shown as undisclosed recipients. The subject is “Monday, March 7 2022”. The message contains inconsistent service/branding references.

Observed indicators

• Unexpected fax notification that the recipient may not be expecting.
• Sender address uses an anonymized attacker-style domain rather than a clearly verified business domain.
• Recipient information is hidden/undisclosed in the supplied sample.
• A prominent “Print & Preview Here” link encourages the recipient to follow an unexpected link.
• Branding references appear inconsistent, which can indicate impersonation or a copied template.
• SPF, DKIM and DMARC results are not available in the supplied header excerpt; therefore, this report does **not** claim that those checks failed.

Link review

Displayed link text: **Print & Preview Here**
Supplied destination: **https://attacker.example.com/**

Assessment: the destination is an anonymized placeholder in the training sample. Its live reputation has not been checked. An unexpected document/fax link is suspicious and should not be opened unless independently verified through a trusted channel.

Classification

Risk: HIGH – PHISHING (training classification). The unexpected fax lure, suspicious sender context, inconsistent branding and link-based call to action collectively indicate phishing behaviour.

5. Sample 2 – Fake Voicemail Notification

Sample description

The sample claims that a voicemail has been received and asks the recipient to download an attached HTML file to listen to the message. The supplied sender is **noreply@target.example.com**, and the subject refers to voice-message access. The sample body includes a caller ID and a short message duration to make the notification appear legitimate.

Header observations

The supplied authentication results state: **SPF=fail**, **DKIM=none** and **DMARC=fail**. The supplied Received-SPF text also says that the sender domain does not designate the listed client IP as a permitted sender. The IP shown, **192.0.2.1**, and the domain are anonymized/example values, so these results describe the supplied sample only.

Observed indicators

• Unexpected voicemail notification with a request to download an HTML attachment.
• SPF failure and DMARC failure in the supplied header results.
• DKIM signature is reported as absent (dkim=none).
• The HTML attachment name resembles a message/audio file but is an HTML document, not a normal audio recording.
• The sample evidence shows an Outlook-like sign-in page, which raises concern about credential harvesting. The attachment was not executed during this review.

Attachment and link review

Supplied attachment name: **audio_john.doe@target.example.com_file.htm**
The extension **.htm** indicates an HTML file. The training evidence indicates that it presents an Outlook-style sign-in page. The sample also references an anonymized redirect/domain. No live redirect chain was independently tested. Do not open unexpected HTML attachments or enter credentials into pages reached from email attachments.

Classification

Risk: HIGH – PHISHING (training classification). The authentication failures, missing DKIM signature, unexpected HTML attachment and sign-in-page behaviour are strong warning signs.

6. Comparative Findings

Check	Sample 1 – Fake Fax	Sample 2 – Voicemail
Sender context	attack@attacker.example.com; undisclosed recipients	noreply@target.example.com
Authentication	Not available in supplied excerpt	SPF fail; DKIM none; DMARC fail
Main lure	New fax / preview document	New voicemail / download to listen
Link or file	Unexpected preview link	Unexpected .htm attachment; sign-in page shown in sample evidence
Risk	High – Phishing	High – Phishing

7. Risk Classification Guide

Safe: No meaningful suspicious indicators are found and the sender/message can be independently verified. “Safe” does not mean risk-free.
Suspicious: One or more warning signs exist, but the available evidence is insufficient to confidently label the message phishing.
Phishing: Multiple strong indicators suggest an attempt to steal credentials, deliver malicious content, impersonate a trusted service or manipulate the recipient into unsafe action.

Both samples are classified as High Risk – Phishing because of the combined indicators described above. The classification is limited to the provided training evidence.

8. Prevention Guidelines – Employee Do’s and Don’ts

Do:
• Verify unexpected fax, voicemail, invoice, password-reset or document notifications through a known official website or phone number.
• Check the full sender address and compare the domain with the organization’s genuine domain.
• Hover over links (without clicking) to inspect the destination; when in doubt, navigate to the service manually using a saved bookmark.
• Treat unexpected HTML, HTM, ZIP, ISO, executable or macro-enabled attachments as suspicious.
• Report suspicious messages to the IT/security team using the organization’s approved process.
• If credentials were entered, immediately contact IT/security and change the password through the official site; enable MFA where available.

Don’t:
• Do not click unexpected “View”, “Preview”, “Listen”, “Download” or “Verify” links.
• Do not open unexpected attachments to find out what they contain.
• Do not enter passwords, MFA codes or personal information on a page opened from an unsolicited email.
• Do not trust a message only because it contains a logo, familiar name, caller ID or professional formatting.
• Do not forward suspicious attachments to colleagues; report the message safely instead.

9. Incident Response – If a User Interacts

If a user clicks a suspicious link, opens an attachment or submits credentials: (1) stop interacting with the page/file; (2) disconnect the affected device from the network if malware execution is suspected and follow company policy; (3) notify the IT/SOC team immediately; (4) preserve the original email and report details; (5) change exposed passwords through the official service and revoke active sessions where appropriate; and (6) allow the security team

to scan and investigate the device. Do not delete evidence unless instructed by the response team.

10. Conclusion

The two training samples demonstrate common phishing themes: unexpected document or voicemail notifications, urgency/curiosity, suspicious links or attachments, and attempts to imitate familiar services. Sample 2 additionally includes explicit SPF and DMARC failures and no DKIM signature in the supplied header results. Employees should verify unexpected requests independently, avoid opening untrusted attachments, and report suspicious messages promptly.

11. Evidence and Project Files

The project folder should contain the following supporting files (use the filenames below if they match the files you created):

- 01_Email_Samples/Sample_01_Phishing_Email.txt
- 01_Email_Samples/Sample_02_Voicemail_Email.txt
- 02_Header_Analysis/Sample_01_Header_Analysis.txt
- 02_Header_Analysis/Sample_02_Header_Analysis.txt
- 03_Link_Analysis/Sample_01_Link_Analysis.txt
- 03_Link_Analysis/Sample_02_Link_Analysis.txt
- 04_Evidence/Sample_01_Fake_Fax/ – fake fax screenshot
- 04_Evidence/Sample_02_Voicemail/ – voicemail screenshot
- 05_Final_Report/ – this final report

12. References and Acknowledgement

Public educational sample repository: https://github.com/autinerd/phishing-mail-examples. The samples are used for educational analysis and attribution. The repository owner is not the report author. Sample names, domains, IP addresses and other identifiers may be anonymized or altered in the source material.

Header review references: Google Admin Toolbox – Messageheader; MXToolbox – Email Headers. These are reference tools; no confidential email data should be submitted without authorization.

Disclaimer: This document is an educational project report, not a forensic certification, legal opinion, or confirmation that any real organization or domain is malicious. Findings are based only on the sample content and header information described in this report.